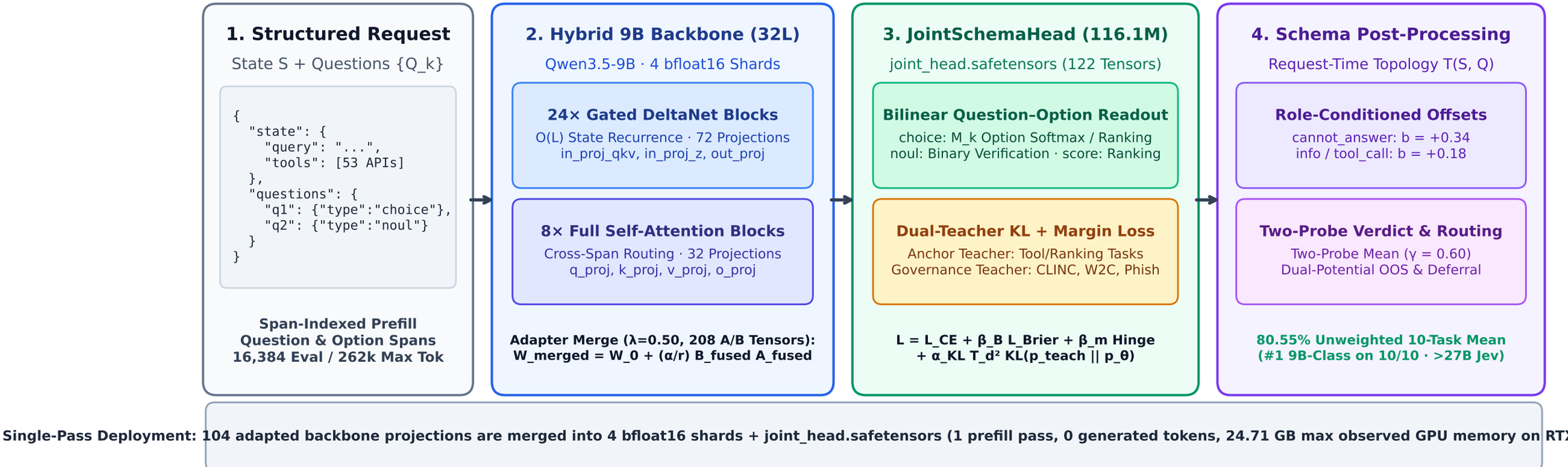

1. Structured Request
State S + Questions {Q_k}
{
"state": {
"query": "...",
"tools": [53 APIs]
},
"questions": {
"q1": {"type":"choice"},
"q2": {"type":"noul"}
}
}
Span-Indexed Prefill
Question & Option Spans
16,384 Eval / 262k Max Tok
2. Hybrid 9B Backbone (32L)
Qwen3.5-9B · 4 bfloat16 Shards
24× Gated DeltaNet Blocks
O(L) State Recurrence · 72 Projections
in_proj_qkv, in_proj_z, out_proj
8× Full Self-Attention Blocks
Cross-Span Routing · 32 Projections
q_proj, k_proj, v_proj, o_proj
Adapter Merge (λ=0.50, 208 A/B Tensors):
W_merged = W_0 + (α/r) B_fused A_fused
3. JointSchemaHead (116.1M)
joint_head.safetensors (122 Tensors)
Bilinear Question-Option Readout
choice: M_k Option Softmax / Ranking
noul: Binary Verification · score: Ranking
Dual-Teacher KL + Margin Loss
Anchor Teacher: Tool/Ranking Tasks
Governance Teacher: CLINC, W2C, Phish
L = L_CE + β_B L_Brier + β_m Hinge
+ α_KL T_d² KL(p_teach || p_θ)
4. Schema Post-Processing
Request-Time Topology T(S, Q)
Role-Conditioned Offsets
cannot_answer: b = +0.34
info / tool_call: b = +0.18
Two-Probe Verdict & Routing
Two-Probe Mean (γ = 0.60)
Dual-Potential OOS & Deferral
80.55% Unweighted 10-Task Mean
(#1 9B-Class on 10/10 · >27B Jev)
Single-Pass Deployment: 104 adapted backbone projections are merged into 4 bfloat16 shards + joint_head.safetensors (1 prefill pass, 0 generated tokens, 24.71 GB max observed GPU memory on RTX PRO 6000 Blackwell).